2 In a diesel engine, a fixed amount of gas undergoes a cycle of four stages. The cycle is shown in Fig. 2.1.

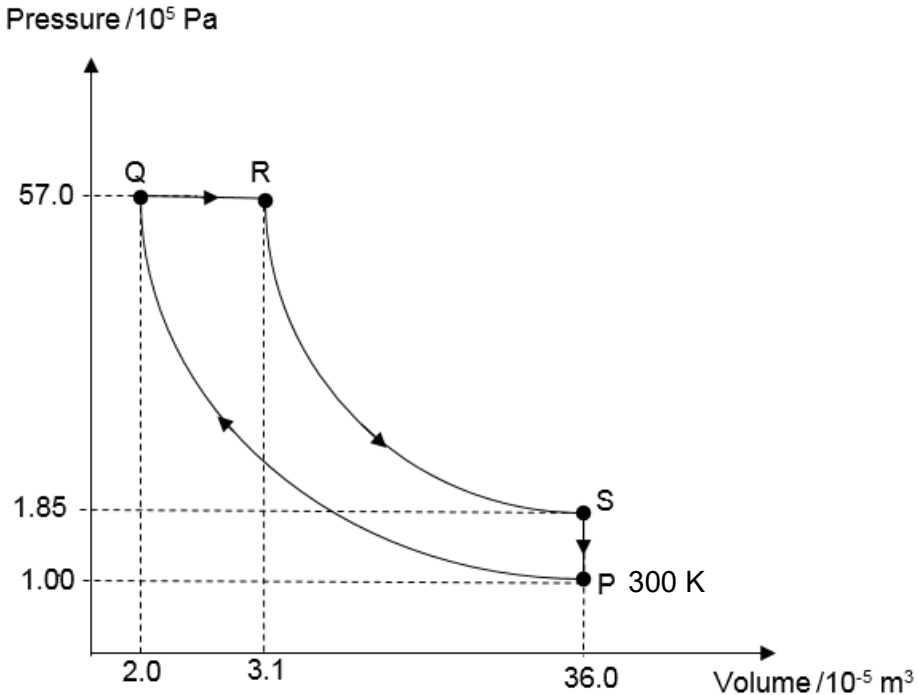

Fig. 2.1 (not to scale)

The four stages are

P → Q : compression with a rise in temperature and pressure,
Q → R : expansion at constant pressure while fuel is being burnt,
R → S : expansion with a drop in both temperature and pressure,
S → P : decrease in pressure at constant volume.

Some numerical values of temperature, pressure and volume are given on Fig. 2.1.

(a) Using Fig. 2.1, calculate the work done by the gas during the stages

(i) Q → R,

work done = …….J [1]

(ii) S → P.

work done = …….J [1]

(b) Using your answers in **(a)**, complete Fig. 2.2 for the four stages of the cycle.

Stage of cycle	heat supplied to gas /J	work done on gas /J	increase in internal energy of the system /J
P → Q	0	235	
Q → R	246		
R → S	0	–333	
S → P			

Fig. 2.2 [4]

(c) Assuming that the gas is ideal, calculate the temperature of the gas at point Q.

temperature = …… K [2]